WHAT SURVIVES THE ETH REFUTATIONS? A MAP OF FINE-GRAINED HYPOTHESES AND CONDITIONAL LOWER BOUNDS

VAIBHAV TULSYAN

Abstract. In October 2026 OpenAI and Anthropic claimed deterministic subexponential-time algorithms for #k-SAT, refuting the Exponential Time Hypothesis (ETH) and with it the Strong ETH. This note records, row by row, which fine-grained hypotheses and which published conditional lower bounds are affected, assuming the claimed algorithms are correct. We classify 33 hypotheses and 27 conditional theorems as refuted, as having lost their premise, or as surviving; a fourth label marks the few theorems whose conclusion is now contradicted by an explicit algorithm. The map is organised by one distinction: the new algorithms handle k-CNF formulas of bounded width, and, through the width reduction of Calabro, Impagliazzo and Paturi, CNFs of any width with at most $n \cdot 2^{o((\log n)^{1/4}/\sqrt{\log\log n})}$ clauses (a short derivation of ours from the explicit rate of the Anthropic manuscript), but nothing is known for CNF-SAT with polynomially many clauses of unbounded width. Many of the best-known lower bounds (treewidth optimality, edit distance, Fréchet distance, LCS, diameter approximation) are proved from that "CNF-SETH" and survive with their premise intact. Gap-ETH and the ETH-based proof of the Parameterized Inapproximability Hypothesis lose their force; Almost-ETH, OV, 3SUM, APSP and the Set Cover Conjecture are untouched. We list the few genuinely new algorithms that follow. A refutation removes premises; it does not by itself give algorithms, and nothing here is practical. Everything is conditional on the claimed refutations, which are under review.

1. Introduction

1.1. **The two refutations.** The Exponential Time Hypothesis (ETH) of Impagliazzo and Paturi [37, 38] defines s_k, for $k \geq 3$, as the infimum of the δ such that k-SAT on n variables is solvable in time $O(2^{\delta n})$, and asserts $s_k > 0$ for $k \geq 3$; the Strong ETH (SETH) is $s_\infty = \lim_k s_k = 1$ [38]. In October 2026 two manuscripts appeared, by OpenAI [54] and by Anthropic [21], each claiming a deterministic algorithm that counts the satisfying assignments of a k-CNF in subexponential time. Both come with Lean 4 formalisations of their top-level statements. These are *claimed* results, still under review. We write [A] and [O] for them, and use them in the following forms.

Imported Theorem 1.1 ([A], Main Theorem, eq. (1))**.** *There is an absolute constant C and a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in time*

$$2^{C\cdot\min\{k^2 n \log\log n/\sqrt{\log n},\; km/\sqrt{\log m}\}}.$$

Imported Theorem 1.2 ([O], main theorem)**.** *For each fixed k there is a deterministic algorithm that counts the models of a k-CNF with n variables and m clauses in $(n+m)^{O_k(1)}2^{O_k(n\log\log n/\sqrt{\log n})}$ bit operations and polynomial space, and in $2^{O_{k,D}(n/\sqrt{\log n})}$ when $m \leq Dn$.*

Two features matter below. First, Imported Theorem 1.1 is *uniform in k*: C is absolute, so the bound is subexponential in n as long as $k = o((\log n)^{1/4}/\sqrt{\log\log n})$. [O] states no dependence on k, so it is not known to be uniform. Uniformity is irrelevant for ETH and SETH: each engine gives $s_k = 0$ for every fixed k, hence $s_\infty = 0$ ([A], Corollary 5; the reduction from 3-SAT to k-SAT is the deterministic sparsification of [39]). It matters only for formulas whose width grows with n (Section 2). Second, both algorithms are deterministic and count, so they refute the randomized, counting and nondeterministic variants of ETH and SETH *for k-CNFs* at once ([A], Corollaries 5 and 6 and the remark after Corollary 5).

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1.2. **The direction of implication.** A conditional lower bound "$H \Rightarrow$ problem X needs time T" is proved by a reduction *from* a problem covered by H *to* X. Refuting H removes the premise; it says nothing about X. New algorithms for X arise only when there is a reduction in the other direction, from X to k-SAT with a linear number of variables (the SERF framework of [39], and [A], Corollary 7), or from more general theorems inside the manuscripts that the top-level statements do not expose (our own sparse results, Section 5). Most rows below therefore lose a premise and gain nothing. We say so row by row.

1.3. **Statuses.** Throughout, "refuted" means *refuted assuming the claimed refutations.*

- **refuted**: the hypothesis itself is false.
- **premise refuted**: a theorem $H \Rightarrow X$ with H false; nothing here contradicts X.
- **survives**: we know of no consequence of the refutations that touches the statement. For a theorem this means that its premise, as actually used in the proof, still stands.

The additional label **conclusion refuted** marks a theorem $H \Rightarrow X$ whose conclusion X is contradicted by an explicit algorithm. One row is a theorem whose premise is the *negation* of ETH, so that the theorem now **fires** (Section 4.1); a few rows are **mixed** and are split in the reason column.

1.4. **Sources.** The map was compiled from the primary sources: for every row tagged ^{L} in Tables 1 and 2 the wording was checked against the abstract or full text of the cited source (arXiv, ECCC, DROPS, journal, or authors' copy). Rows tagged ^{F} are folklore conjectures without a single primary statement; ^{I} marks our own inference, and ^{O} our own results. In an earlier pass, nine rows rested on secondary sources or on sources we had not re-read (ETH itself; the CIP duality; Williams' Max-2-CSP algorithm; Feige's hypothesis; Pătraşcu–Williams, twice; Cai–Juedes; Chen–Huang–Kanj–Xia; Marx's treewidth paper). All nine were re-verified against the primary text or abstract for this note, so no row now rests on a secondary source. Theorem numbers refer to the versions named in the bibliography.

2. Bounded width versus unbounded width

2.1. **Three facts.**

(F1) $s_k = 0$ for every k. This refutes ETH, SETH, #ETH, #SETH, $\oplus$SETH and NSETH in their k-CNF forms.

(F2) *Local witnesses.* A problem with complexity parameter n whose yes-instances have an $O(n)$-bit witness checkable by an $O(n)$-gate circuit, or by bounded-width clauses on $O(n)$ variables, gets deterministic time $2^{\delta n}$ for every $\delta > 0$ by Tseitin's transformation; counts follow when the encoding is parsimonious. This is the SERF route [39]; [A], Corollary 7, uses it with a cardinality circuit of $O(n)$ gates.

(F3) *The width frontier* (Proposition 2.1 below). CNFs of arbitrary width are reached only when the number of clauses is barely superlinear.

2.2. **The width reduction and the density frontier.** Calabro, Impagliazzo and Paturi [15] relate CNF-SAT of bounded density to k-SAT by Schuler's width reduction: while some clause C has more than k literals, pick k of them, C', and branch on "C' is satisfied" (replace C by C') versus "C' is falsified" (set its k literals to false). The branches are disjoint, so model counts add up, and the leaves are k-CNFs. This counting form is [60, Lemma 6.1]; with constant k it gives #CNF-SAT with $m \le cn$ clauses in time $2^{\varepsilon n}$ for all $c, \varepsilon > 0$, from either engine [60, Corollary 6.2]. The uniformity of Imported Theorem 1.1 lets k grow, and gives a little more.

Proposition 2.1 (our derivation from Imported Theorem 1.1)**.** *Put $L(n) = (\log n)^{1/4}/\sqrt{\log\log n}$. Assume Imported Theorem 1.1. For every function f with $f(n) = o(L(n))$ there is a deterministic algorithm that counts the models of CNF formulas of arbitrary width with n variables and at most $n \cdot 2^{f(n)}$ clauses in time $2^{o(n)}$.*

Proof. Let $m \le Dn$ with $\log D = f(n)$, and fix $k \ge 2$, to be chosen. Run the width reduction. Along a root-to-leaf path every "satisfied" branch removes one long clause, so there are at most m of them, and every "falsified" branch fixes k variables, so there are $r \le n/k$ of them. The leaves with exactly r falsified branches number at most $\binom{m+r}{r}$ [15, proof of Lemma 5], so there are at most $(n/k+1)\binom{m+\lfloor n/k \rfloor}{\lfloor n/k \rfloor} \le (n+1)\big(e(Dk+1)\big)^{n/k}$ leaves. Each leaf is a k-CNF with $n' \le n$ variables and at most m clauses. If $n' \le \sqrt{n}$, count by brute force

in time $2^{\sqrt{n}}\,\mathrm{poly}(m)$. Otherwise $\log n' \geq \frac{1}{2}\log n$, so $\log\log n'/\sqrt{\log n'} \leq 2\log\log n/\sqrt{\log n}$ for n large, and Imported Theorem 1.1 counts the leaf in time 2^{sn} with $s = 2Ck^2\log\log n/\sqrt{\log n}$. The total time is at most

$$\mathrm{poly}(n,m)\cdot 2^{\,(n/k)\,(f(n)+\log(k+1)+2)\,+\,sn\,+\,\sqrt{n}}.$$

Take $k = \lceil\sqrt{\max\{f(n),1\}\cdot L(n)}\,\rceil$. Then $k\to\infty$, $f(n)/k \leq \sqrt{f(n)/L(n)} \to 0$ and $(\log k)/k \to 0$, while $k/L(n)\to 0$ gives $s = 2C(k/L(n))^2 \to 0$. Every term of the exponent is $o(n)$, and $\mathrm{poly}(n,m) = 2^{o(n)}$ since $m \leq n^{1+o(1)}$. □

Remark 2.2 (what does not follow).

(a) The clause-count branch of Imported Theorem 1.1 does worse here: it needs $km = o(n\sqrt{\log m})$, which allows only $m \leq n(\log n)^{1/2-o(1)}$.

(b) Through [O] only constant density is reached explicitly, since its O_k-constants are not stated; a diagonal argument gives some unbounded but non-explicit density.

(c) For $m = n^{1+\Omega(1)}$, or $m \geq n\cdot 2^{(\log n)^{1/4+\Omega(1)}}$, nothing follows: the leaf count forces $k \gtrsim \log(m/n)$, where neither engine is subexponential in n. Hence the statement

CNF-SETH: for every $\varepsilon > 0$, CNF-SAT with n variables and m clauses of unbounded width is not solvable in time $2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$

survives, together with its counting, parity and nondeterministic forms.

CNF-SETH is exactly the premise that many famous reductions use. Lokshtanov, Marx and Saurabh [45] assume that "SAT cannot be solved in $(2-\varepsilon)^n m^{O(1)}$ time" and stress that they need no sparsification; Backurs and Indyk [7] reduce from CNF-SAT with M clauses to edit distance; Bringmann's "SETH′" [11] is "no $O^*((2-\delta)^N)$ algorithm for CNF-SAT"; Roditty and Vassilevska Williams [56, Theorem 1] conclude "CNF-SAT on n variables is in $O^*((2-\delta)^n)$"; and the moderate-dimension Orthogonal Vectors conjecture follows from CNF-SETH by Williams' split-and-list reduction with $d = m$. What dies is everything that genuinely goes through sparsification to $O(n)$ clauses, or through bounded width.

3. Table A: hypotheses

Table 1 lists the hypotheses. "Gives algorithms" is noted in the reason column when a refutation also yields an explicit faster algorithm for the problem named.

Table 1: Hypotheses, assuming the claimed refutations.

	Hypothesis (source)	Status	Reason
A1	ETH: $s_k > 0$ for $k \geq 3$ [37, 38]^{L}	refuted	Either engine; 3-SAT and k-SAT in $2^{o(n)}$.
A2	SETH: $s_\infty = 1$ [38]; randomized algorithms allowed in [24]^{L}	refuted	(F1); [O] suffices despite non-uniformity in k.
A3	Density-SETH: CNF of any width with $\leq cn$ clauses needs $2^{(1-\varepsilon)n}$ [15]; [52, Conj. 2]^{L}	refuted	Counting width reduction [60, Cor. 6.2].
A4	CNF-SETH: CNF-SAT $\notin 2^{(1-\varepsilon)n}\,\mathrm{poly}(m)$, unbounded width [45, 7, 11, 56]^{L}	survives	Proposition 2.1 and Remark 2.2: reach ends at $n2^{o(L(n))}$ clauses.
A5	Super-SETH: for every unbounded f, k-SAT $\notin 2^{n(1-f(k)/k)}$ [62]^{L}	refuted	It implies SETH.
A6	#ETH [29, p. 21:5]^{L}	refuted	[A], Corollary 6; [O]. Gives #3-SAT in $2^{o(n)}$.
A7	#SETH, $\oplus$SETH for k-CNF; equivalent to k-$\oplus$Hitting Set and k-$\oplus$Set Cover hardness [24, Thm. 1.3]^{L}	refuted	(F1).
A8	#CNF-SETH, $\oplus$CNF-SETH ($\mathrm{poly}(m)$ clauses)^{I}	survives	As A4.
A9	NSETH: $\forall\varepsilon\,\exists k$, k-TAUT $\notin$ NTIME$[2^{(1-\varepsilon)n}]$ [16]^{L}	refuted	A deterministic algorithm is a nondeterministic one ([A], after Corollary 5).
A10	NSETH for CNF with $\mathrm{poly}(m)$ clauses, or for poly-size circuits [52, Conj. 3]^{L}	survives	As A4.
A11	MA-SETH, AM-SETH	refuted (2016)	Williams [63]: MA proofs in c^n time, $c < 2$, for #SAT of sublinear-depth circuits; "strongly refute" both^{L}. For k-CNF also by (F1).

	Hypothesis (source)	Status	Reason
A12	Gap-ETH: no $2^{o(n)}$ (randomized) algorithm distinguishes satisfiable 3-CNFs from those not $(1-\varepsilon)$-satisfiable [30, 50], as quoted in [17]^{L}	refuted	An exact deterministic $2^{o(n)}$ decider distinguishes the cases. [49] also records Gap-ETH as false.
A13	ETH for Circuit-SAT of linear size; VSP-Circuit-SAT of size cn [24, Thm. 1.1(v)]^{L}	refuted	Tseitin gives $O(n)$ variables; gives algorithms.
A14	Circuit-, NC- and BP-SETH [4]^{L}	survives	Superlinear-size representations.
A15	FPT $\neq$ M[1] (miniaturised CNF of size $k \log n$) [32, §3.5]. Chen–Grohe: M[1] = FPT iff 3-SAT $\in$ DTIME($2^{o^{\mathrm{eff}}(n)}$) [20]^{L}	refuted	The iff needs an effective little-o; [A] (rate with absolute C) and [O] at $k=3$ supply one. Gives algorithms for M[1]-complete problems.
A16	FPT $\neq$ W[1]	survives	ETH implies it, not conversely; [20, §6] list "ETH $\Leftrightarrow$ FPT $\neq$ W[1]?" as open^{L}.
A17	Low-dimension OVC: $\forall\varepsilon\,\exists c$, OV with $d = c\log n$ $\notin n^{2-\varepsilon}$ ("LDOVC" of [34])^{L}	survives	SETH support gone (it was via sparsification [12, Lemma 2.1]); [A] notes it does not refute OV.
A18	Moderate-dimension OVC: no $n^{2-\varepsilon}\,\mathrm{poly}(d)$ [34]^{L}	survives	Implied by CNF-SETH with $d = m$ [2, Lemma 1]; premise intact.
A19	k-OV, low and moderate dimension^{I}	survives	As A17, A18.
A20	Hitting Set conjecture [6]^{L}	survives	Not known to follow from SETH; implies OVC.
A21	Set Cover Conjecture: $\forall\varepsilon\,\exists k$, Set Cover with sets of size $\leq k \notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, n the universe [24, Thm. 1.2]^{L}	survives	Not implied by SETH even before 2026; the natural witness has $\Theta(n\log n)$ bits, so (F2) does not apply.
A22	k-Hitting Set, k-Set Splitting, k-NAE-SAT need $2^{(1-\varepsilon)n}$ for large k, each equivalent to SETH [24, Thm. 1.1]^{L}	refuted	Equivalent to SETH; directly, they are k-CNFs on the universe. Gives algorithms.
A23	3SUM conjecture^{F}	survives	No SAT premise; see B26.
A24	APSP; combinatorial BMM; OMv^{F}	survives	Unrelated to SAT.
A25	k-Clique in $n^{\omega k/3-\varepsilon}$; min-, exact- and zero-weight k-Clique^{F}	survives	Unrelated.
A26	(ℓ,k)-Hyperclique hypothesis [44, Hyp. 1.4]^{L}	survives	Its refutation would give dense Max-k-SAT in $2^{(1-\varepsilon)n}$, which A27 leaves open; no converse from SAT.
A27	Dense Max-3-SAT $\notin 2^{(1-\varepsilon)n}$; Max-2-SAT, Max-Cut not below $2^{\omega n/3}$ (algorithm: [64])^{F}	survives	A counter over $\Theta(n^2)$ clauses has $\Theta(n^2)$ variables.
A28	Subset Sum $\notin 2^{(1/2-\varepsilon)n\,\mathrm{F}}$	survives	Weights are not local.
A29	Basic QSETH (CNF-SAT) [13]; its k-CNF variant^{L}	mixed	CNF form survives (as A4); k-CNF variant refuted [60, Table 2, row 23].
A30	PIH [48]^{L}	survives	Unproved again: both proofs, from Gap-ETH and from ETH [35], lose their premises.
A31	ETH for End-of-a-Line [57]^{L}	survives	A different, query-type hypothesis.
A32	Feige's random 3-SAT hypothesis [33]^{L}	survives	About polynomial-time average-case refutation.
A33	Almost-ETH: $\forall\gamma>0$, 3-SAT $\notin O(2^{n^{1-\gamma}})$ [49]^{L}	survives	Both engines give $2^{n/(\log n)^{\Theta(1)}} = 2^{n^{1-o(1)}}$, not $2^{n^{1-\gamma}}$.

Remarks. Row A11 was refuted before 2026 and is listed for completeness. Row A15 is the one place where an *effective* rate matters: Chen and Grohe's miniaturisation isomorphism maps SUBEPT to FPT, and the explicit rates of both manuscripts make the conclusion M[1] = FPT unconditional modulo the refutations. Row A33 was proposed by Manurangsi in the 27 October 2026 revision of [49] because of the refutations; its comment states that the earlier versions' ETH and Gap-ETH premises "are now known to be false".

4. Table B: conditional theorems

Table 2: Conditional theorems "$H \Rightarrow X$", assuming the claimed refutations.

	Theorem (source)	Status	Reason
Parameterised and exact exponential			
B1	Treewidth DP optimality: no $(2-\varepsilon)^{\mathrm{tw}}$ IS, $(3-\varepsilon)^{\mathrm{tw}}$ DomSet, $(2-\varepsilon)^{\mathrm{tw}}$ Max-Cut, $(3-\varepsilon)^{\mathrm{tw}}$ OCT, $(q-\varepsilon)^{\mathrm{tw}}$ q-Col, $(2-\varepsilon)^{\mathrm{tw}}$ Partition into Triangles [45]^{L}	survives	The premise as proved is CNF-SETH (A4), with $m^{O(1)}$ clauses; no sparsification is used. No faster DP follows.

	Theorem (source)	Status	Reason
B2	Cut&Count pathwidth bounds: no $(3-\varepsilon)^{\mathrm{pw}}$ CVC, FVS, Steiner Tree; no $(4-\varepsilon)^{\mathrm{pw}}$ CDS, CFVS, COCT; Exact k-Leaf Spanning Tree [26, Thm. 1.3, App. D]^{L}	survives	Every reduction of App. D starts from CNF-SAT with n variables and m clauses: CNF-SETH. (Their Thm. 1.4 is in B21.)
B3	Pătraşcu–Williams: k-DomSet in $O(n^{k-\varepsilon})$ for some $k \geq 3 \Rightarrow$ CNF-SAT in $2^{\delta n}\,\mathrm{poly}(m)$, $\delta < 1$ [55, Thm. 2.1]^{L}	survives	CNF form.
B4	Pătraşcu–Williams: d-SUM on N numbers of $O(d \log N)$ bits, $d < N^{0.99}$, in $N^{o(d)} \Rightarrow k$-SAT in $2^{o(n)}$ [55, §5]^{L}	premise refuted	Conclusion of the contrapositive is now true; d-SUM bound open.
B13	ETH $\Rightarrow$ no $2^{o(n)}$ IS, Clique, VC, k-Col, DomSet on n vertices, any density [39, 46]^{L}	conclusion refuted	[A], Corollary 7: deterministic $O(2^{\delta n})$ for every δ, any number of edges, with counts by size and of k-colourings.
B14	ETH $\Rightarrow$ k-VC $\notin 2^{o(k)}\,\mathrm{poly}(n)$ (from B13, as $k \leq n$; cf. [14])^{L}	conclusion refuted^{I}	$2k$-vertex LP kernel [25, §2.5], then Corollary 7 (Section 5). Cai–Juedes: Max-SNP-complete problems have $2^{o(k)}p(n)$ algorithms iff 3-SAT $\in$ DTIME$(2^{o(n)})$.
B15	ETH $\Rightarrow$ no $2^{o(n)}$ Max-Cut, Max-k-SAT with $O(n)$ constraints^{F}	conclusion refuted^{I}	Unweighted, or $O(1)$-bit weights; Section 5. Polynomial weights: under $(\mathrm{H_A})$, or by the white-box extension of [O] to $\mathbb{Q}$-algebra weights in [8] (not a consequence of the top-level theorem of [O]). Dense: A27.
B16	ETH $\Rightarrow$ no $2^{o(n)}$ HC, HP, TSP, Steiner, FVS, CVC, CDS, Max-Leaf on sparse graphs [46, Thm. 3.3]	conclusion refuted under $(\mathrm{H_A})^{\mathrm{O}}$	[58] and our notes [59]; on dense graphs only the premise is refuted.
B19	ETH $\Rightarrow$ k-Clique $\notin f(k)n^{o(k)}$ [18, Thm. 5.5]^{L}	premise refuted	$2^{\delta n}$ does not contradict it at small k.
B20	ETH $\Rightarrow$ binary CSP $\notin f(G)n^{o(k/\log k)}$, k the treewidth [51]; #ETH $\Rightarrow$ #k-matching (bipartite) $\notin f(k)n^{o(k/\log k)}$ [23]^{L}	premise refuted	Parameter $k \ll n$. On a class of bounded-degree graphs of unbounded treewidth, Nguyen [53] decides binary CSP in $f(G)n^{O(k/\sqrt{\log k})}$ via the network engine of [A], contradicting the literal (unconditional) form of Marx's Conjecture 1.4 ($n^{o(k)}$); the exponent $o(k/\log k)$ of the theorem is not reached.
B21	ETH $\Rightarrow$ no $2^{o(d \log d)}$ Closest String, Distortion, Disjoint Paths by tw, ... [47]; no $2^{o(p \log p)}$ Cycle Packing [26, Thm. 1.4]^{L}	premise refuted	
B22	ETH $\Rightarrow$ $2^{\Omega(\sqrt{n})}$ planar bounds [46]; $2^{\Omega(n^{1-1/d})}$ for fat-object graphs [28] and Euclidean TSP [27]^{L}	premise refuted	The refutations give only $2^{n/\mathrm{polylog}\,n}$, worse than the upper bounds. Under Almost-ETH the reductions give $2^{n^{1/2-\gamma}}$-type bounds^{I}.
B11	Under the Set Cover Conjecture: Steiner Tree $\notin 2^{\varepsilon t}$ poly, CVC $\notin 2^{\varepsilon t}$ poly, Set Partitioning $\notin 2^{\varepsilon n}\,\mathrm{poly}(m)$, Subset Sum $\notin t^{\varepsilon}\,\mathrm{poly}(n)$ [24, Thm. 1.2]^{L}	survives	A21 intact.
B12	Subset Sum $\notin T^{1-\varepsilon}2^{o(n)}$ [3]^{L}	premise refuted	Reduces from k-SAT; the $T^{1-\varepsilon}\,\mathrm{poly}(n)$ form survives via B11.
Approximation			
B23	Gap-ETH $\Rightarrow$ FPT-inapproximability of k-Clique, k-DomSet, Biclique, induced matching, DkS [17]^{L}	premise refuted	Partly re-proved under W[1] $\neq$ FPT (survives): constant-factor k-Clique [43], $k^{1/H(k)}$ for k-Clique [41], any $F(k)$ for k-DomSet [42]. The ETH parts (KLM's $T(k)n^{o(k)}$; the $k^{o(1)}$ bounds discussed in [19]) lose their premise. No W[1] re-proof found for the others.
B24	ETH-hardness of approximation: 2-CSP to $\lvert V\rvert^{1-o(1)}$, Directed Steiner Network to $k^{1/4-o(1)}$ [31]; DkS [49]; best Nash [10]; free games [1]^{L}	mixed	DkS to $2^{(\log n)^{1-\varepsilon}}$ re-proved under Almost-ETH in [49] (v3): survives. [31, 10, 1]: premise refuted; quasi-polynomial reductions plausibly give weaker bounds under Almost-ETH^{I}.
B25	ETH $\Rightarrow$ PIH [35]; near-optimal $f(k)n^{o(k)}$-type bounds for approximate k-Clique and CSP under ETH [36]^{L}	premise refuted	See A30.

	Theorem (source)	Status	Reason
Problems in P			
B5	Edit distance $\notin n^{2-\varepsilon}$ [7]; also from NC-SETH [4]^{L}	survives	Reduction from CNF-SAT with M clauses: CNF-SETH.
B6	LCS, DTW: Most-OV with threshold, $n^{2-\varepsilon}\,\mathrm{poly}(d) \Rightarrow k$-LCS $\notin n^{k-\varepsilon}$ [2, Thm. 1, Conj. 1]; binary LCS, edit distance with costs, DTW [12]^{L}	survives	[2, Lemma 1]: Conj. 1 follows from Max-CNF-SAT hardness with $\mathrm{poly}(m)$ clauses. [12] is stated under k-SETH but proved from OVH, which CNF-SETH implies.
B7	Fréchet distance $\notin n^{2-\varepsilon}$; no 1.001-approximation in $(nm)^{1-\delta}$ [11]^{L}	survives	Stated from SETH′: no $O^*((2-\delta)^N)$ CNF-SAT.
B8	Diameter: no $(3/2-\varepsilon)$-approximation in $O(m^{2-\varepsilon})$, undirected unweighted [56, Thm. 1]^{L}	survives	Stated with conclusion "CNF-SAT in $O^*((2-\delta)^n)$". Later k-OV-based tradeoffs not checked here.
B9	Dynamic problems under SETH (#SSR, SubConn, SCC, ...) [5]^{L}	survives^{I}	Stated via k-SAT with $O(n)$ clauses (sparsification), which is refuted; but the constructions use one node per clause, so the proofs run from CNF-SETH. Triangle, BMM and 3SUM-based bounds untouched.
B10	Results using $d = c\log n$ exactly, stated "under SETH"	premise refuted	They survive if restated under low-dimension OVC (A17).
B26	NSETH $\Rightarrow$ no deterministic fine-grained reduction from SAT to 3SUM, APSP [16]^{L}	premise refuted	The barrier is lifted; no reduction follows.
Counting, quantum, circuits			
B17	#ETH bounds for the Tutte polynomial, the permanent and #IS [29, 22, 9]^{L}	mixed	Conclusion refuted: #2-SAT and #IS on n variables/vertices at any density ([A], Corollaries 6, 7); sparse permanent, #PM, Boolean #CSP, the integer hyperbolas H_q (local encodings; also, independently and earlier, [8]); $T(x,1)$ on sparse graphs under $(\mathrm{H_A})^{\mathrm{O}}$. Dense permanent and the rest: premise refuted.
B18	Quantum-supremacy statements from ETH/SETH [52] and others	mixed	Of 24 rows in [60, Table 2]: 5 refuted, 5 premise refuted, 1 moot, 1 open, 12 survive.
B27	ETH false $\Rightarrow \mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ [65, Thm. 6.1]; see [40]^{L}	fires	Section 4.1.

Rows are grouped by area rather than by number; the numbering follows our working map in [59]. Row B3 is stated in [55] as "Hypothesis 1 implies that CNF-SAT has an improved algorithm"; Section 2 of [55] remarks that the earlier $f(k)n^{o(k)}$ bound for k-DomSet of Chen et al. concerns only k-SAT, which is precisely the distinction of Section 2.

4.1. **Circuit lower bounds.** Williams [65, Theorem 6.1] proved that if ETH is false then E^{NP} has no circuits of linear size; [A] draws this consequence and formalises it in Lean. Modulo the refutations, $\mathrm{E}^{\mathrm{NP}} \not\subseteq \mathrm{SIZE}(O(n))$ is therefore unconditional. In a companion note [61] we track the constants through Williams' argument and the local reductions of Jahanjou, Miles and Viola [40]: assuming the clause-count bound of Imported Theorem 1.1 in a Turing-machine form, there are $\varepsilon > 0$ and $L_0 \in \mathrm{E}^{\mathrm{NP}}$ with no circuits of size $\varepsilon n\sqrt{\log n}$ for infinitely many n. That result is machine-checked in Lean on top of Anthropic's own circuit-lower-bound module (`enp_not_size_n_sqrt_log`; standard axioms; fresh kernel replay), holds only infinitely often, and has a non-explicit ε. It gives nothing at $n\log n$ and nothing for NEXP. Carmosino et al. [16] note that disproving NSETH gives circuit lower bounds; for the k-CNF form refuted here this is subsumed by the above.

5. The new algorithms

The refutations give few explicit algorithms beyond k-SAT itself. We list all we know of, separating what needs only the formalised top-level statements from what needs internal results of [A].

5.1. **From the main theorems alone.**

(i) k-SAT, #k-SAT, $\oplus k$-SAT and k-TAUT; #CNF-SAT of any width with $O(n)$ clauses (either engine), and with $n2^{o(L(n))}$ clauses ([A]; Proposition 2.1); linear-size Circuit-SAT.

(ii) [A], Corollary 7: for every $\delta > 0$, deterministic $O(2^{\delta n})$ algorithms that count the independent sets, cliques, vertex covers and dominating sets of each size s, and the proper k-colourings, of a graph on n vertices with any number of edges. Decision for IS, Clique, VC and k-Colouring was implied before by [39]; Dominating Set and the counts are new in [A].

(iii) k-Hitting Set, k-Set Splitting, k-NAE-SAT, #2-SAT, sparse permanent, #PM, Boolean #CSP, and the integer Potts and flow hyperbolas of the Tutte plane on sparse graphs (local encodings, (F2)).

(iv) *Vertex Cover in* $2^{o(k)}+\mathrm{poly}(n)$ (our observation). The LP kernel leaves at most $2k$ vertices in polynomial time [25, §2.5]; Corollary 7 then counts vertex covers of size k in $2^{2\delta k}$ for every δ. A single algorithm in time $2^{O(k \log\log k/\sqrt{\log k})} + \mathrm{poly}(n)$ follows by running the explicit n-bound of Imported Theorem 1.1 on the $O(k)$-variable 3-CNF of the proof of Corollary 7. This contradicts the conclusion of Cai–Juedes' ETH bound (B14), consistently with their own equivalence.

(v) *Sparse Max-Cut and Max-k-SAT, unweighted* (our observation). For $m = O(n)$ constraints with $O(1)$-bit weights, encode each constraint by a forced indicator variable, add an $O(m)$-gate adder and a comparison with a target t, and apply Tseitin: a 3-CNF with $O(n)$ variables whose models are the assignments of value exactly t. Corollary 6 counts them for each t, giving the exact optimum and the number of solutions of each value in $2^{\delta n}$. Weights up to W cost $O(m \log W)$ gates, so polynomial weights are not covered by the main theorems. Bhattacharjee [8] obtains them by running the algorithm of [O] itself over truncated polynomial rings, a white-box extension.

(vi) M[1]-complete problems in FPT time (A15).

5.2. **From internal results of [A].** Our other results assume three internal statements of [A] (a network-to-clique reduction, an algorithm for cliques of all levels, and the existence of its "pieces"), called ($\mathrm{H_A}$) in [58, 60]. They are not consequences of the top-level theorems.

(vii) *Sparse Hamiltonicity* [58]: assuming ($\mathrm{H_A}$), the number of Hamiltonian cycles of a graph with n vertices and $O(n)$ edges in $O(2^{\delta n})$ for every δ, and in $2^{O(\max(m,n)/\sqrt{\log n})}$ in general; hence decision, search, Hamiltonian paths, sparse TSP with polynomial weights, directed cycles and ATSP. These theorems are machine-checked in Lean on top of the development of [A], relative to ($\mathrm{H_A}$).

(viii) *Exact quantum amplitudes* [60]: assuming ($\mathrm{H_A}$), amplitudes of Clifford+T circuits with G two-qubit gates in $2^{O(G/\sqrt{\log G})}$, machine-checked relative to ($\mathrm{H_A}$); marginals and exact sampling on paper.

(ix) *Further sparse problems, on paper only* [59]: Steiner Tree, FVS, Connected VC and DS, Max-Leaf; the line $T(x, 1)$ of the Tutte plane (forests counted by size) on sparse graphs, which is the line Brand, Dell and Roth [9] show #ETH-hard; sparse Max-Cut with polynomial weights. All are conditional on ($\mathrm{H_A}$) and not formalised. For Max-Cut and Max-k-SAT with polynomial weights, Bhattacharjee [8] gives an earlier, independent route that replaces ($\mathrm{H_A}$) by the claim that the steps of [O] remain valid over a commutative $\mathbb{Q}$-algebra (unrefereed, not formalised).

5.3. **What gets no algorithm.** Nothing parameterised by treewidth or pathwidth, except binary CSP on one class of bounded-degree graphs, with exponent $O(\mathrm{tw}/\sqrt{\log \mathrm{tw}})$ [53] (B20); no problem in P (edit distance, LCS, Fréchet, diameter, OV, 3SUM, APSP, k-Clique); Set Cover by universe size; Subset Sum; dense Max-CSP; dense Hamiltonicity; planar problems (where $2^{O(\sqrt{n})}$ is already better).

6. Reading the map by area

6.1. **Exact and parameterised algorithms.** The treewidth and pathwidth bounds (B1, B2) are the clearest case of a premise that survives because of how it is stated. Lokshtanov, Marx and Saurabh build, from a CNF formula with n variables and m clauses, a graph of size polynomial in $n + m$ and treewidth roughly $n/\log_2 c$ (the precise form varies by problem), so a $(c-\varepsilon)^{\mathrm{tw}}$ algorithm would decide CNF-SAT in time $2^{(1-\delta)n}\,\mathrm{poly}(m)$. Nothing in the reduction requires bounded width or few clauses, and Remark 2.2 shows that the new algorithms reach only $m \le n2^{o(L(n))}$. To refute B1 one would need CNF-SAT with, say, n^2 clauses of width up to n in time $2^{(1-\delta)n}$; the width reduction would then need $k \approx \log n$, where the bound of Imported Theorem 1.1 is $2^{\Theta(n(\log n)^{3/2}\log\log n)}$, far above 2^n.

The ETH-based bounds for parameterised problems (B19–B22) lose their premise, but are not contradicted: a $2^{\delta n}$ algorithm says nothing about $f(k)n^{o(k)}$ when $k \ll n$. The two exceptions are structural. Chen and

Grohe's equivalence (A15) is an "if and only if", so its other direction now yields M[1] = FPT. And Vertex Cover parameterised by solution size has a linear kernel, so the parameter and the number of vertices coincide after preprocessing, and the $2^{\delta n}$ algorithm of [A] becomes a $2^{o(k)}$ algorithm (B14). Whether W[1] $\neq$ FPT (A16) is unaffected: Chen and Grohe list its equivalence with ETH as open, and the refutations settle ETH without touching it.

6.2. **Approximation.** Gap-ETH is refuted by any exact $2^{o(n)}$ algorithm, so it falls with ETH; the gap does not help. The FPT-inapproximability results of Chalermsook et al. (B23) therefore lose their premise, and so do the ETH-hardness results of Dinur–Manurangsi, Braverman–Ko–Weinstein and Aaronson–Impagliazzo–Moshkovitz (B24) and the ETH proof of PIH (B25). Two kinds of result survive. The first are the re-proofs under W[1] $\neq$ FPT: constant-factor k-Clique (Lin), $k^{1/H(k)}$-factor k-Clique (Karthik–Khot) and $F(k)$-factor k-DomSet (Karthik–Laekhanukit–Manurangsi). The second are results whose reductions are only mildly exponential, so that a weaker hypothesis suffices: Manurangsi's Densest-k-Subgraph bound, now stated under Almost-ETH (A33). The refutations give 3-SAT in $2^{n/(\log n)^c}$ for every constant $c < 1/2$, which is $2^{n^{1-o(1)}}$ but not $2^{n^{1-\gamma}}$, so Almost-ETH stands. We expect other quasi-polynomial-size reductions (Nash, free games, 2-CSP) to give weaker but non-trivial bounds under Almost-ETH; we have not checked this ([I]).

6.3. **Problems in P.** Here the map is almost entirely "survives". Edit distance, Fréchet distance and diameter are stated from CNF-SAT with poly(m) clauses (B5, B7, B8); LCS and DTW go through Most-OV or OV with dimension $d = m$ (B6), which CNF-SETH implies. What is lost is the derivation of the *low-dimension* OV conjecture ($d = c \log n$) from SETH, which uses sparsification to $O(n)$ clauses (A17, B10). OVC itself, 3SUM, APSP and the hyperclique hypothesis were never consequences of ETH. The dynamic lower bounds of Abboud and Vassilevska Williams (B9) are a borderline case: stated through sparsification, but by our reading proved by constructions that also run from CNF-SETH. The barrier of Carmosino et al. (B26), that NSETH forbids deterministic fine-grained reductions from SAT to 3SUM and APSP, disappears for k-SAT; no such reduction is known.

6.4. **Counting and quantum.** Counting statements behave like decision ones: #ETH and the k-CNF form of #SETH fall (A6, A7), and lower bounds proved by local reductions on sparse instances are contradicted by explicit algorithms (B17): Corollaries 6 and 7 of [A] count independent sets and models of 2-CNFs at any density, and local encodings handle sparse permanents, perfect matchings and the integer hyperbolas of the Tutte plane. The non-local cases (Hamiltonian cycles, forests) need (H_A). Dense permanents, and the Tutte plane on dense graphs, are untouched. For the quantum statements (B18) the same split appears: sparse Clifford+T circuits fall under (H_A), while dense or average-case conjectures survive [60, §6].

7. Summary

Status	Table A (33)	Table B (27)	Total
refuted	12	–	12
conclusion refuted	–	4 (B13–B16)	4
fires	–	1 (B27)	1
premise refuted	–	10	10
survives	20	9	29
mixed	1 (A29)	3 (B17, B18, B24)	4

Table 3. Counts per status. Refuted: A1–A3, A5–A7, A9, A11–A13, A15, A22. Premise refuted: B4, B10, B12, B19–B23, B25, B26. Survives: A4, A8, A10, A14, A16–A21, A23–A28, A30–A33, B1–B3, B5–B9, B11.

Three points stand out. First, the most-cited SETH lower bounds, for treewidth DP and for edit distance, Fréchet distance, LCS and diameter, survive with an intact premise, because they are proved from CNF-SAT with polynomially many clauses; what dies is "k-SETH $\Rightarrow$ low-dimension OVC" and anything that genuinely needs sparsification. Second, Gap-ETH is refuted, so the Gap-ETH programme in parameterised inapproximability and the ETH proof of PIH lose their premises; the parts re-proved under W[1] $\neq$ FPT

survive. Third, the frontier of the new algorithms in clause density is sharp in an unusual way: $n \cdot 2^{o(L(n))}$ clauses through the explicit rate of [A], constant density through [O], and nothing at $n^{1+\Omega(1)}$.

8. Limitations

(a) *Everything is conditional* on the claimed refutations [21, 54], whose expert review is pending. If they fail, every "refuted" becomes "open" and every "premise refuted" reverts to a valid conditional theorem; Proposition 2.1 and the observations of Section 5 lose their proofs. Results marked (H_A) need internal statements of [A] that are not part of its formalised top level.

(b) *Direction.* A refutation removes premises. Except through (F2) and the network theorems, it gives no algorithm, and "premise refuted" is never evidence that a conclusion is false.

(c) *Sources.* Every row was checked against its primary source; the nine rows that earlier rested on secondary sources were re-verified for this note (Section 1), and no row is now secondary. Rows marked ^{F} have no single primary statement; rows marked ^{I} are our inferences and are argued only in the reason column. B8 omits later diameter tradeoffs. The map is not exhaustive: [A] reports more than 800 papers with results proved only under ETH or a hypothesis implying it.

(d) *Proofs.* Proposition 2.1 and the observations (iv)–(v) of Section 5 are short written arguments, produced with AI assistance and not machine-checked or reviewed by a human expert. Only the results so labelled in Section 5 are machine-checked, and only relative to their stated hypotheses.

(e) *Nothing practical.* [A] notes that its bound does not beat 2^n for 3-CNFs with $5n$ clauses unless $n > 10^{10\,000}$. All algorithms here are galactic.

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References

1. Scott Aaronson, Russell Impagliazzo, and Dana Moshkovitz, *AM with multiple Merlins*, Proc. 29th IEEE Conference on Computational Complexity (CCC), 2014, arXiv:1401.6848, pp. 44–55.
2. Amir Abboud, Arturs Backurs, and Virginia Vassilevska Williams, *Tight hardness results for LCS and other sequence similarity measures*, Proc. 56th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2015, arXiv:1501.07053, titled "Quadratic-time hardness of LCS and other sequence similarity measures", pp. 59–78.
3. Amir Abboud, Karl Bringmann, Danny Hermelin, and Dvir Shabtay, *SETH-based lower bounds for subset sum and bicriteria path*, Proc. 30th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2019, arXiv:1704.04546. Journal version: ACM Trans. Algorithms **18** (2022), no. 1, pp. 41–57.
4. Amir Abboud, Thomas Dueholm Hansen, Virginia Vassilevska Williams, and Ryan Williams, *Simulating branching programs with edit distance and friends, or: a polylog shaved is a lower bound made*, Proc. 48th Annual ACM Symposium on Theory of Computing (STOC), 2016, arXiv:1511.06022, pp. 375–388.
5. Amir Abboud and Virginia Vassilevska Williams, *Popular conjectures imply strong lower bounds for dynamic problems*, Proc. 55th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2014, arXiv:1402.0054, pp. 434–443.
6. Amir Abboud, Virginia Vassilevska Williams, and Joshua Wang, *Approximation and fixed parameter subquadratic algorithms for radius and diameter in sparse graphs*, Proc. 27th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2016, arXiv:1506.01799, pp. 377–391.
7. Arturs Backurs and Piotr Indyk, *Edit distance cannot be computed in strongly subquadratic time (unless SETH is false)*, SIAM J. Comput. **47** (2018), no. 3, 1087–1097, Preliminary version in STOC 2015; arXiv:1412.0348.
8. Deep Bhattacharjee, *Subexponential weighted counting and exact optimisation*, Hexagon preprint hexagon:2610.00594v1, 29 October 2026, https://hexagonmath.org/abs/2610.00594, 2026.
9. Cornelius Brand, Holger Dell, and Marc Roth, *Fine-grained dichotomies for the Tutte plane and Boolean #CSP*, Algorithmica **81** (2019), no. 2, 541–556, Preliminary version in IPEC 2016; arXiv:1606.06581.
10. Mark Braverman, Young Kun Ko, and Omri Weinstein, *Approximating the best Nash equilibrium in $n^{o(\log n)}$-time breaks the exponential time hypothesis*, Proc. 26th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2015, ECCC TR14-092, pp. 970–982.
11. Karl Bringmann, *Why walking the dog takes time: Fréchet distance has no strongly subquadratic algorithms unless SETH fails*, Proc. 55th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2014, arXiv:1404.1448, pp. 661–670.
12. Karl Bringmann and Marvin Künnemann, *Quadratic conditional lower bounds for string problems and dynamic time warping*, Proc. 56th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2015, arXiv:1502.01063, pp. 79–97.

13. Harry Buhrman, Subhasree Patro, and Florian Speelman, *The quantum strong exponential-time hypothesis*, arXiv:1911.05686 (v2). Published as "A framework of quantum strong exponential-time hypotheses", Proc. 38th STACS, LIPIcs **187** (2021), 19:1–19:19, 2019.
14. Liming Cai and David Juedes, *On the existence of subexponential parameterized algorithms*, J. Comput. System Sci. **67** (2003), no. 4, 789–807.
15. Chris Calabro, Russell Impagliazzo, and Ramamohan Paturi, *A duality between clause width and clause density for SAT*, Proc. 21st Annual IEEE Conference on Computational Complexity (CCC), 2006, pp. 252–260.
16. Marco L. Carmosino, Jiawei Gao, Russell Impagliazzo, Ivan Mihajlin, Ramamohan Paturi, and Stefan Schneider, *Nondeterministic extensions of the strong exponential time hypothesis and consequences for non-reducibility*, Proc. 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS), 2016, ECCC TR15-148, pp. 261–270.
17. Parinya Chalermsook, Marek Cygan, Guy Kortsarz, Bundit Laekhanukit, Pasin Manurangsi, Danupon Nanongkai, and Luca Trevisan, *From Gap-ETH to FPT-inapproximability: clique, dominating set, and more*, Proc. 58th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2017, arXiv:1708.04218. Journal version: SIAM J. Comput. **49** (2020), no. 4, 772–810, pp. 743–754.
18. Jianer Chen, Xiuzhen Huang, Iyad A. Kanj, and Ge Xia, *Strong computational lower bounds via parameterized complexity*, J. Comput. System Sci. **72** (2006), no. 8, 1346–1367.
19. Yijia Chen, Yi Feng, Bundit Laekhanukit, and Yanlin Liu, *Simple combinatorial construction of the $k^{o(1)}$-lower bound for approximating the parameterized k-clique*, Proc. 2025 Symposium on Simplicity in Algorithms (SOSA), 2025, arXiv:2304.07516, pp. 263–280.
20. Yijia Chen and Martin Grohe, *An isomorphism between subexponential and parameterized complexity theory*, SIAM J. Comput. **37** (2007), no. 4, 1228–1258, Preliminary version in CCC 2006; ECCC TR06-011.
21. Claude (Anthropic), *A $2^{o(n)}$ time algorithm for 3-SAT*, Preprint dated October 26, 2026. Paper and Lean formalisation in the repository `https://github.com/anthropics/formal-math`, folder `eth`, 2026.
22. Radu Curticapean, *Block interpolation: a framework for tight exponential-time counting complexity*, Inform. and Comput. **261** (2018), 265–280, Preliminary version in ICALP 2015.
23. Radu Curticapean and Dániel Marx, *Complexity of counting subgraphs: only the boundedness of the vertex-cover number counts*, Proc. 55th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2014, arXiv:1407.2929, pp. 130–139.
24. Marek Cygan, Holger Dell, Daniel Lokshtanov, Dániel Marx, Jesper Nederlof, Yoshio Okamoto, Ramamohan Paturi, Saket Saurabh, and Magnus Wahlström, *On problems as hard as CNF-SAT*, ACM Trans. Algorithms **12** (2016), no. 3, Art. 41, Preliminary version in CCC 2012. Theorem numbers as in arXiv:1112.2275v3.
25. Marek Cygan, Fedor V. Fomin, Łukasz Kowalik, Daniel Lokshtanov, Dániel Marx, Marcin Pilipczuk, Michał Pilipczuk, and Saket Saurabh, *Parameterized algorithms*, Springer, 2015.
26. Marek Cygan, Jesper Nederlof, Marcin Pilipczuk, Michał Pilipczuk, Johan M. M. van Rooij, and Jakub Onufry Wojtaszczyk, *Solving connectivity problems parameterized by treewidth in single exponential time*, Proc. 52nd IEEE Symposium on Foundations of Computer Science (FOCS), 2011, Journal version in ACM Trans. Algorithms 18 (2022), no. 2, pp. 150–159.
27. Mark de Berg, Hans L. Bodlaender, Sándor Kisfaludi-Bak, and Sudeshna Kolay, *An ETH-tight exact algorithm for Euclidean TSP*, Proc. 59th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2018, arXiv:1807.06933, pp. 450–461.
28. Mark de Berg, Hans L. Bodlaender, Sándor Kisfaludi-Bak, Dániel Marx, and Tom C. van der Zanden, *A framework for ETH-tight algorithms and lower bounds in geometric intersection graphs*, Proc. 50th Annual ACM Symposium on Theory of Computing (STOC), 2018, arXiv:1803.10633, pp. 574–586.
29. Holger Dell, Thore Husfeldt, Dániel Marx, Nina Taslaman, and Martin Wahlén, *Exponential time complexity of the permanent and the Tutte polynomial*, ACM Trans. Algorithms **10** (2014), no. 4, Art. 21.
30. Irit Dinur, *Mildly exponential reduction from gap 3SAT to polynomial-gap label-cover*, Electronic Colloquium on Computational Complexity, TR16-128, 2016.
31. Irit Dinur and Pasin Manurangsi, *ETH-hardness of approximating 2-CSPs and directed Steiner network*, 9th Innovations in Theoretical Computer Science Conference (ITCS 2018), LIPIcs, vol. 94, 2018, arXiv:1805.03867, pp. 36:1–36:20.
32. Rodney G. Downey and Dimitrios M. Thilikos, *Confronting intractability via parameters*, Comput. Sci. Rev. **5** (2011), no. 4, 279–317, Section numbers as in arXiv:1106.3161.
33. Uriel Feige, *Relations between average case complexity and approximation complexity*, Proc. 34th Annual ACM Symposium on Theory of Computing (STOC), 2002, pp. 534–543.
34. Jiawei Gao, Russell Impagliazzo, Antonina Kolokolova, and Ryan Williams, *Completeness for first-order properties on sparse structures with algorithmic applications*, Proc. 28th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2017, ECCC TR16-053. Journal version: ACM Trans. Algorithms **15**, no. 2, pp. 2162–2181.
35. Venkatesan Guruswami, Bingkai Lin, Xuandi Ren, Yican Sun, and Kewen Wu, *Parameterized inapproximability hypothesis under ETH*, Proc. 56th Annual ACM Symposium on Theory of Computing (STOC), 2024, arXiv:2311.16587. Journal version: J. ACM **72** (2025), no. 5, pp. 24–35.
36. ———, *Almost optimal time lower bound for approximating parameterized clique, CSP, and more, under ETH*, Proc. 57th Annual ACM Symposium on Theory of Computing (STOC), 2025, arXiv:2404.08870, pp. 2136–2144.
37. Russell Impagliazzo and Ramamohan Paturi, *Complexity of k-SAT*, Proc. 14th Annual IEEE Conference on Computational Complexity (CCC), 1999, pp. 237–240.
38. ———, *On the complexity of k-SAT*, J. Comput. System Sci. **62** (2001), no. 2, 367–375.
39. Russell Impagliazzo, Ramamohan Paturi, and Francis Zane, *Which problems have strongly exponential complexity?*, J. Comput. System Sci. **63** (2001), no. 4, 512–530.

40. Hamid Jahanjou, Eric Miles, and Emanuele Viola, *Local reductions*, Automata, Languages, and Programming (ICALP 2015), Part I, Lecture Notes in Comput. Sci., vol. 9134, Springer, 2015, Numbering as in arXiv:1311.3171. Journal version: "Local reduction", Inform. and Comput. **261** (2018), 281–295, pp. 749–760.
41. Karthik C. S. and Subhash Khot, *Almost polynomial factor inapproximability for parameterized k-clique*, 37th Computational Complexity Conference (CCC 2022), LIPIcs, vol. 234, 2022, arXiv:2112.03983, pp. 6:1–6:21.
42. Karthik C. S., Bundit Laekhanukit, and Pasin Manurangsi, *On the parameterized complexity of approximating dominating set*, Proc. 50th Annual ACM Symposium on Theory of Computing (STOC), 2018, arXiv:1711.11029. Journal version: J. ACM **66** (2019), no. 5, pp. 1283–1296.
43. Bingkai Lin, *Constant approximating k-clique is W[1]-hard*, Proc. 53rd Annual ACM Symposium on Theory of Computing (STOC), 2021, arXiv:2102.04769, pp. 1749–1756.
44. Andrea Lincoln, Virginia Vassilevska Williams, and Ryan Williams, *Tight hardness for shortest cycles and paths in sparse graphs*, Proc. 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2018, arXiv:1712.08147, pp. 1236–1252.
45. Daniel Lokshtanov, Dániel Marx, and Saket Saurabh, *Known algorithms on graphs of bounded treewidth are probably optimal*, Proc. 22nd Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2011, arXiv:1007.5450. Journal version: ACM Trans. Algorithms **14** (2018), no. 2, pp. 777–789.
46. ———, *Lower bounds based on the Exponential Time Hypothesis*, Bull. Eur. Assoc. Theor. Comput. Sci. EATCS **105** (2011), 41–72.
47. ———, *Slightly superexponential parameterized problems*, SIAM J. Comput. **47** (2018), no. 3, 675–702, Preliminary version in SODA 2011; arXiv:1902.08723.
48. Daniel Lokshtanov, M. S. Ramanujan, Saket Saurabh, and Meirav Zehavi, *Parameterized complexity and approximability of directed odd cycle transversal*, Proc. 31st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2020, arXiv:1704.04249, pp. 2181–2200.
49. Pasin Manurangsi, *Almost-polynomial ratio Almost-ETH-hardness of approximating densest k-subgraph*, arXiv:1611.05991v3, 27 October 2026. Versions 1–2 (2016), titled "Almost-polynomial ratio ETH-hardness of approximating densest k-subgraph", appeared in Proc. 49th STOC (2017), 954–961, 2026.
50. Pasin Manurangsi and Prasad Raghavendra, *A birthday repetition theorem and complexity of approximating dense CSPs*, arXiv:1607.02986, 2016.
51. Dániel Marx, *Can you beat treewidth?*, Theory Comput. **6** (2010), 85–112.
52. Tomoyuki Morimae and Suguru Tamaki, *Fine-grained quantum computational supremacy*, arXiv:1901.01637 (v4, 2019); published in Quantum Inf. Comput. **19** (2019), no. 13&14, 1089–1115, 2019.
53. Troy Tri Nguyen, *A sublinear treewidth exponent for binary constraint satisfaction*, Hexagon preprint hexagon:2611.00004v1, 2 November 2026, https://hexagonmath.org/abs/2611.00004, 2026.
54. OpenAI, *The Exponential Time Hypothesis is false*, Preprint dated 17 October 2026, revised 25 October 2026. Paper and Lean formalisation in the repository https://github.com/openai/math, 2026.
55. Mihai Pătraşcu and Ryan Williams, *On the possibility of faster SAT algorithms*, Proc. 21st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA), 2010, pp. 1065–1075.
56. Liam Roditty and Virginia Vassilevska Williams, *Fast approximation algorithms for the diameter and radius of sparse graphs*, Proc. 45th Annual ACM Symposium on Theory of Computing (STOC), 2013, arXiv:1207.3622, titled "Approximating the diameter of a graph", pp. 515–524.
57. Aviad Rubinstein, *Settling the complexity of computing approximate two-player Nash equilibria*, Proc. 57th Annual IEEE Symposium on Foundations of Computer Science (FOCS), 2016, arXiv:1606.04550, pp. 258–265.
58. Vaibhav Tulsyan, *Counting Hamiltonian cycles in sparse graphs in subexponential time, assuming the new #SAT algorithms*, Hexagon preprint hexagon:2610.00672v1, October 2026; source and Lean development at https://github.com/xennygrimmato/eth-hamiltonicity, 2026.
59. ———, *eth-hamiltonicity: Lean 4 formalisation and research notes*, Public repository https://github.com/xennygrimmato/eth-hamiltonicity, Lean sources as of commit https://github.com/xennygrimmato/eth-hamiltonicity/tree/ca17a3762bb21ddbe5942eeac9f29fa58123bb42, 2026.
60. ———, *Exact amplitudes of sparse quantum circuits in subexponential time, assuming the new #SAT algorithms*, Manuscript, October 2026; `docs/quantum-note.pdf` in the repository https://github.com/xennygrimmato/eth-hamiltonicity, 2026.
61. ———, *Superlinear circuit lower bounds for* E^{NP}*, assuming the new #SAT algorithms*, Manuscript, November 2026; `docs/circuit-note.pdf` in the repository https://github.com/xennygrimmato/eth-hamiltonicity, 2026.
62. Nikhil Vyas and Ryan Williams, *On super strong ETH*, J. Artificial Intelligence Res. **70** (2021), 473–495, Preliminary version in SAT 2019, Lecture Notes in Comput. Sci. 11628, 406–423.
63. Richard Ryan Williams, *Strong ETH breaks with Merlin and Arthur: short non-interactive proofs of batch evaluation*, 31st Conference on Computational Complexity (CCC 2016), LIPIcs, vol. 50, 2016, arXiv:1601.04743, pp. 2:1–2:17.
64. Ryan Williams, *A new algorithm for optimal 2-constraint satisfaction and its implications*, Theoret. Comput. Sci. **348** (2005), no. 2–3, 357–365.
65. ———, *Improving exhaustive search implies superpolynomial lower bounds*, SIAM J. Comput. **42** (2013), no. 3, 1218–1244, Conference version in STOC 2010. Theorem numbers as in the author's version https://people.csail.mit.edu/rrw/improved-algs-lbs2.pdf.